Persona 1: The Nominal Match (Baseline Retrieval)

Buyer Profile: Corporate Travel Director for a mid-sized multinational looking for a predictable, standard mission fit.

Ready-to-Copy Prompt:

"Hello. Our company is looking to purchase a new business jet. We need an aircraft that can fly nonstop between São Paulo (GRU) and Manaus (MAO), roughly 1,500 nautical miles, with 6 passengers and standard luggage. Our destination runway in secondary hubs is typically around 4,500 feet. We require an enclosed lavatory and an acquisition budget between $10M and $12M. Which aircraft fits our requirements?"

Evaluation Target: Tests accurate multi-attribute filtering (Range ≥ 1,500 nm, Runway ≤ 4,500 ft, Seats ≥ 6, Budget $10M–$12M).

Expected AI Behavior: Recommends a light jet such as the **Embraer Phenom 300E** or **Cessna Citation CJ4 Gen2**, highlighting runway performance and mission comfort.

Persona 2: The Out-of-Bounds Mission (Negative / Refusal Test)

Buyer Profile: High-Net-Worth individual with unrealistic physical and financial expectations.

Ready-to-Copy Prompt:

"Hi, I want to buy a private jet to fly nonstop from São Paulo to Lisbon (approx. 4,200 nm) with 8 passengers on board. However, I need to operate out of an unpaved grass strip on my farm that is only 2,200 feet long, and my total acquisition budget is capped at $8 million. What aircraft can fulfill this mission?"

Evaluation Target: Tests hallucination resistance, physical impossibility detection, and graceful refusal.

Expected AI Behavior: Explains that no business jet exists that can fly 4,200 nm from a 2,200 ft unpaved strip within an $8M budget. Deconstructs the physical constraints (runway length vs. MTOW for intercontinental jets) and suggests chartering or a hub-and-spoke strategy.

Persona 3: The Multi-Match Trade-Off (Comparative Synthesis)

Buyer Profile: CFO of an investment fund evaluating competitive super-midsize jets.

Ready-to-Copy Prompt:

"We are evaluating super-midsize jets for transcontinental flights across the Americas. Non-negotiables: NBAA IFR range of at least 3,200 nm with 4 passengers, a true stand-up flat-floor cabin (minimum 6'0" height), and modern avionics. We are considering models like the Praetor 600, Challenger 3500, and Citation Longitude. Could you present a side-by-side comparison of the options in your database, analyzing range, cabin volume, runway requirements, and pricing trade-offs?"

Evaluation Target: Tests side-by-side comparative synthesis, structured tables, and trade-off articulation rather than picking a single winner.

Expected AI Behavior: Outputs a clear comparison matrix across the candidates, highlighting specific differentiators (e.g., Praetor 600's range and runway agility vs. Challenger 3500's cabin width).

Persona 4: The Contradictory Requirements (Conflict Resolution)

Buyer Profile: Tech startup founder with mutually exclusive budget and performance targets.

Ready-to-Copy Prompt:

"I want to purchase a brand-new jet that can comfortably fly 9 passengers nonstop from New York to London (approx. 3,500 nm) at High-Speed Cruise (Mach 0.85+). However, our finance board has set an absolute hard cap of $6.5 million for the brand-new purchase price. Which model matches this?"

Evaluation Target: Tests conflict identification and active constraint relaxation.

Expected AI Behavior: Flags the conflict immediately. Explains that new aircraft capable of Mach 0.85 transatlantic missions cost upwards of $20M+, whereas $6.5M new only accesses entry-level Very Light Jets (VLJs). Proposes relaxing the budget, looking at pre-owned inventory, or reducing range/passenger requirements.

Persona 5: The Vague Buyer (Ambiguity & Elicitation)

Buyer Profile: Executive Assistant asking on behalf of a C-level executive without technical details.

Ready-to-Copy Prompt:

"Hi there! Our CEO mentioned we need to acquire a private jet soon for travel between our corporate offices in Brazil and the United States. Could you tell me which airplane we should purchase?"

Evaluation Target: Tests proactive inquiry and information gathering. The assistant must not recommend an aircraft prematurely.

Expected AI Behavior: Acknowledges the broad goal and asks structured clarifying questions: specific city pairs (e.g., Miami vs. New York vs. Los Angeles), passenger count, frequency of flights, airfield constraints, and budget expectations.

Persona 6: The "Hot & High" Operator (Critical Runway Physics)

Buyer Profile: Chief Pilot for an enterprise operating in mountainous or high-elevation territories.

Ready-to-Copy Prompt:

"We need an aircraft that can regularly operate out of high-elevation airfields such as Toluca, Mexico (elevation ~8,460 ft) in ISA+15 conditions, departing with enough fuel and 5 passengers to fly 1,800 nm. The landing runway at our regional destination is 4,100 ft. Which aircraft in the handbook have the engine flat-rating and runway performance to execute this without severe payload or fuel restrictions?"

Evaluation Target: Tests operational depth beyond brochure sea-level performance.

Expected AI Behavior: Filters for aircraft with strong thrust-to-weight ratios, high engine flat-ratings, and low V-speeds, warning against models with high field length penalties at density altitude.

Persona 7: The TCO & Efficiency Buyer (Operating Economics)

Buyer Profile: Operations Director focused strictly on hourly flight economics and fuel efficiency.

Ready-to-Copy Prompt:

"We fly 450 hours per year on domestic flights under 1,000 nm with 3 to 4 executives. Capital for purchase is available, but our main KPI is minimizing Direct Operating Cost (DOC) per flight hour, specifically fuel burn and engine reserve maintenance. We require an enclosed lavatory and a cruising speed of at least 400 ktas. Which aircraft provides the lowest hourly operating cost for this profile?"

Evaluation Target: Tests economic optimization over cabin prestige or maximum range.

Expected AI Behavior: Identifies light jets with low fuel burns (e.g., single-pilot certified light jets) and compares their hourly DOC against larger midsize alternatives, calculating annual cost variance over 450 hours.

Persona 8: Ergonomics & Cabin Comfort (Human Factors Focus)

Buyer Profile: Family Office Advisor purchasing for a tall principal and family with substantial baggage.

Ready-to-Copy Prompt:

"Our principal is 6'4" (1.93 m) tall and travels with family members, requiring ample baggage capacity for sports gear and luggage (minimum 100 cu ft). Required range is moderate, around 2,000 nm. Non-negotiables: a flat-floor cabin with minimum 6'0" ceiling height, low cabin altitude (under 6,000 ft at FL410), and a fully enclosed aft lavatory. Speed and range are secondary to comfort. What models fit?"

Evaluation Target: Tests extraction of interior dimensional and environmental specifications (cabin height, flat floor vs. trenched aisle, baggage volume, cabin altitude).

Expected AI Behavior: Excludes light jets with dropped aisles or low headroom regardless of range; selects aircraft engineered with true flat floors and low cabin pressure altitudes (e.g., Praetor 500/600 or Citation Longitude).

Persona 9: Fleet Commonality & Synergy (Fleet Expansion)

Buyer Profile: Flight Operations Manager for an existing fleet operator seeking step-up synergy.

Ready-to-Copy Prompt:

"Our flight department currently operates two Embraer Phenom 300Es. Due to expanding executive routes into North America and Europe, we need a step-up aircraft with at least 3,500 nm range. We value OEM fleet commonality, crew type-rating transition ease, and leveraging our existing Embraer Executive Care (EEC) support infrastructure. What is the logical transition aircraft in your database, and how does it integrate with our operations?"

Evaluation Target: Tests context-aware brand continuity, avionics philosophy transition, and maintenance ecosystem valuation.

Expected AI Behavior: Recommends the **Praetor 600**, explaining operational commonalities (fly-by-wire cockpit philosophy, customer support synergy, maintenance network) while highlighting the transoceanic capability jump.

Persona 10: Payload-Range Boundary Case (Full Fuel vs. Full Payload)

Buyer Profile: Sports and entertainment logistics manager carrying maximum weight over long distances.

Ready-to-Copy Prompt:

"We need to transport 8 passengers with heavy luggage (225 lbs per person, total payload 1,800 lbs) on a 2,400 nm mission facing 40-knot winter headwinds, requiring standard NBAA IFR reserves. Many light-to-midsize jets advertise ranges of 2,500+ nm. Can they actually fly this with full seats and bags without a payload-range penalty? Please identify aircraft whose Max Payload Range or Full-Fuel Payload supports this specific mission."

Evaluation Target: Tests understanding of Maximum Takeoff Weight (MTOW), Maximum Zero Fuel Weight (MZFW), and payload-range trade-offs (brochure ferry range vs. full-payload mission range).

Expected AI Behavior: Distinguishes between ferry range (with minimal payload) and mission range with 1,800 lbs payload plus headwinds and reserves, eliminating light jets that cannot take full fuel when carrying 8 heavy passengers.
